Persistent Orienteering for Heterogeneous Multi-UAV Environmental Monitoring

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***Abstract*—Persistent environmental monitoring with teams of autonomous drones requires revisiting sites whose criticality regrows over time. Wildfire surveillance is a concrete case: multiple fire fronts ignite, intensify, and demand repeated inspection. This paper formulates the Persistent Orienteering Problem (POP), an extension of the Orienteering Problem in which each node accumulates a time-growing need and a visit reduces that need by an agent- and node-dependent factor. The formulation captures heterogeneous robot teams under energy and recharging limits. This paper then presents a centralized reward-rate policy that scores candidate visits by expected reward per unit travel time and assigns free robots through a sequential-greedy auction in a receding-horizon loop. Numerical simulations with realistic UAV speeds show that, under a matched online replan cadence, the policy reduces the time-averaged monitoring cost by about 10%–20% relative to online TOP (LKH-3), and under a matched depot cadence it remains best on heterogeneous instances (about 5% below depot TOP) while tying depot TOP when inspection quality is uniform. It stays close to assigning each free robot to its best next site. When few robots must cover many sites, the gain over TOP shrinks, but the policy remains competitive. Spatial residual-need maps and GIS wildfire episodes, with full videos linked below, show how the fleet reallocates as need regrows.**

***Index Terms*—persistent monitoring, orienteering problem, multi-robot systems, task allocation, aerial robots**

I. Introduction

Autonomous aerial robots are increasingly deployed to monitor large outdoor environments, agricultural fields, forests, industrial sites, disaster areas. Wildfire monitoring is a representative, high-stakes case: multiple fire foci ignite at different times and locations, each intensifying if left unattended, so a team of drones must keep revisiting affected regions to maintain an up-to-date picture. More generally, the quantity of interest is not static — pollutant concentrations drift, crops develop, structures degrade, fire fronts advance — so information gathered at a location becomes stale over time and a single sweep is insufficient. A team of robots must persistently revisit sites of interest so that accumulated criticality across the environment stays low, under a limited fleet, finite battery life, and heterogeneous sensing capabilities. We keep the formulation and method general and use wildfire monitoring as a concrete, recurring illustration.

The Orienteering Problem (OP) and its Team variant (TOP) [1] model reward-collecting robot routing under a budget constraint, but are fundamentally *one-shot*: rewards are fixed and each node is worth visiting at most once. This mismatches persistent monitoring, where a node's value regrows between visits and a visit only partially reduces accumulated need. Dynamic, criticality-aware extensions exist for single-robot predictive maintenance [2], but the multi-robot, heterogeneous, steady-state case remains open, e.g., a fleet mixing a high-endurance fixed-wing scout with agile multirotors is naturally heterogeneous, each robot more effective at specific sites.

This paper considers a Persistent Orienteering Problem (POP), unifying persistence and team heterogeneity: each node's need grows linearly at a node-specific rate, and the team minimizes long-run time-averaged accumulated need under energy and recharging constraints. We propose a centralized reward-rate policy: a coordinator scores each free agent–node pair by expected reward per unit travel time and assigns robots via a sequential-greedy auction, resolved whenever agents become free, anticipating need growth during travel and favoring the robot that inspects each node best.

The approach differs from prior work in three respects. First, unlike OP/TOP solvers assuming static rewards and a full reset on visit, POP models a partial, agent-dependent multiplicative reset with a steady-state objective. Second, unlike fixed cyclic patrolling, the policy re-allocates robots as the need field changes and an *inspection quality* term varies across the team. Third, unlike one-shot or learning-based planners with high per-replan cost, our controller runs in closed loop with work linear in the team size and the number of nodes per tick, and yields lower monitoring cost.

II. Related Work

The Orienteering Problem (OP) seeks a length-constrained route maximizing rewards collected at visited nodes; surveys cover its variants, solvers, and applications, from single-vehicle OP to team and time-dependent extensions [1], [3]. Classic OP assumes static rewards, whereas monitoring applications demand time-varying profits, spatio-temporal [4] and correlated [5], [6] reward models, uncertain/dynamic rewards [7], and dynamical multirotor vehicle models [8] have all been explored. Closest to ours is the Dynamic Criticality-Aware OP, which schedules a single robot against drifting sensor criticality [2]; POP generalizes this single-robot setting to a multi-robot, heterogeneous, persistent team.

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TABLE I
MODELING ASSUMPTIONS OF RELATED ORIENTEERING FORMULATIONS.

	OP/TOP	Time-var. OP	Patrolling	DCA-OP	POP
Fleet	Team or single	Usually single	Multi-robot	Single robot	Multi-robot
Reward	Static	Time-varying	Regrows	Grows	Regrows
Visit	Collected once	Collected once	Repeated	Repeated	Partial
Sensing	Uniform	Uniform	Uniform	One robot	Heterogeneous
Endurance	Length budget	Length budget	Often omitted	Revisit limits	Energy, recharge
Objective	Collected reward	Collected reward	Idleness	Criticality	Average need

Multi-robot extensions distribute the routing across a fleet. The Team OP has been used for environmental monitoring with service-time rewards [9] and benchmarked for multi-UAV task assignment [10], while risk- and heterogeneity-aware variants include the Team Surviving Orienteers [11] and its matroid generalization for heterogeneous teams [12], as well as coverage-oriented team routing under uncertain reward [13]. A complementary line of work addresses persistent monitoring and multi-robot patrolling, which explicitly model information that regrows between visits: representative approaches compute cyclic routes on graphs [14], analyze simple versus optimal patrolling schedules [15], schedule samples online for spatiotemporal fields [16], and use multi-period OP formulations to capture repeated visits [17]. Finally, learning-based solvers such as transformer and reinforcement-learning policies tackle the (team) OP and its time-window variants [18], [19], and kinematic and Dubins formulations account for motion constraints [20], [21].

Existing formulations assume static or additively time-varying rewards with a *full reset* on visit, or treat robots as homogeneous. None jointly model a multiplicative, partial reset governed by agent- and node-specific inspection quality together with a steady-state, energy-constrained objective. POP fills this gap, and our reward-rate auction is designed to exploit both the persistence dynamics and the team heterogeneity. Table I lists the modeling assumptions of these formulations.

III. PROBLEM FORMULATION

This section formalizes the Persistent Orienteering Problem (POP). We first define the sets and state variables that describe the team of robots and the network of monitoring nodes, then specify the temporal dynamics of the need field (persistence growth and heterogeneous, multiplicative inspection), the energy and recharging model, and finally the steady-state objective that the team must minimize.

A. Sets and State Variables

Let $M = \{1, 2, \ldots, N\}$ be a set of N monitoring nodes (sensors/sites) with fixed positions $x_i \in \mathbb{R}^3$. A team of $|A|$ robots, indexed by $k \in A = \{0, 1, \ldots, |A| - 1\}$, patrols the environment. The robots are partitioned into G disjoint groups; agents in the same group share kinematic and sensing characteristics, so heterogeneity is captured at the group level.

Each node i carries a continuous-time state $u_i(t) \in \mathbb{R}_{\geq 0}$, the *accumulated need* (criticality, or staleness) of that node at time t. The position of robot k at time t is $x_r^k(t) \in \mathbb{R}^3$, and $v_k > 0$ is its flight speed.

B. Temporal Dynamics of the Need Field

The need is governed by two mechanisms: *persistence* (continuous growth) and *inspection* (discrete reduction upon visits).

1) Persistence: Absent any inspection, the need of node i grows linearly with a node-specific drift rate $\alpha_i > 0$, saturating at $u_{\max}$:

$$\dot{u}_i(t) = \alpha_i, \qquad u_i(t) \leq u_{\max}. \tag{1}$$

2) Heterogeneous Inspection: When robot k visits node i at a (discrete) visit event t_v, the need is reduced *multiplicatively* by a factor governed by the agent- and node-specific inspection quality $\beta_{ki} \in (0, 1]$:

$$u_i(t_v^+) = (1 - \beta_{ki})\, u_i(t_v^-). \tag{2}$$

The corresponding collected reward (the removed need) is

$$\Delta u_i(t_v) = \beta_{ki}\, u_i(t_v^-). \tag{3}$$

A high-quality inspection ($\beta_{ki} \to 1$) nearly resets the node, whereas a low-quality inspection removes only a fraction, so revisits by better-suited robots remain valuable.

C. Energy and Recharging

Each robot k has a finite energy budget $e_k(t) \in [0, E_k]$ that drains at a group-dependent rate ρ_k while operating. When e_k falls below a threshold $\underline{e}$, the robot returns to a recharging station x_0 and resumes operation once recharged. This couples routing with endurance and forces the policy to balance monitoring against travel to the depot.

D. Objective

The primary objective is to minimize the *steady-state time-averaged accumulated need* over the whole network:

$$J = \frac{1}{T} \int_0^T \sum_{i \in M} u_i(t)\, dt. \tag{4}$$

This captures the persistent-monitoring goal of keeping global staleness low at all times, in contrast to the one-shot reward maximization of the classical OP. We additionally report, as secondary criteria, the maximum latency (largest inter-visit gap over all nodes), the total collected reward $\sum_{t_v} \beta_{ki} u_i(t_v^-)$, and the reward collected per unit of energy.

IV. METHODOLOGY

This section presents the online controller for POP. The team does not commit to a long open-loop tour. A shared coordinator observes positions x_r^k, free/busy flags, residual energy e_k, committed targets, and the need vector $u(t)$. Robots that already hold a target continue toward it; free robots enter an assignment stage. The coordinator scores every

feasible agent–node pair with the reward-rate index, then runs a sequential-greedy auction. Matching is centralized; local tracking is decentralized. One control tick costs $O(|A|N)$ bids. Fig. 1 summarizes the loop.

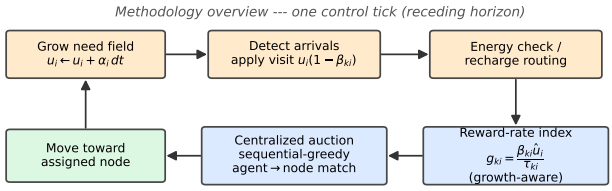


Fig. 1. Control loop: the coordinator grows the need field, applies visits, enforces recharging, scores agent–node pairs by the reward-rate index, runs a sequential auction, and commands motion toward the assigned nodes.

A. Persistence-Aware Reward-Rate Index

Consider robot k at position x_r^k with speed v_k that evaluates node i. Travel to i takes

$$\tau_{ki} = \|x_r^k - x_i\|/v_k. \tag{5}$$

During that interval the need of node i keeps growing. The coordinator therefore predicts the need upon arrival,

$$\hat{u}_i = \min\big(u_i(t) + \alpha_i\,\tau_{ki},\ u_{\max}\big), \tag{6}$$

and defines the reward-rate index as the expected collected reward per unit time:

$$g_{ki} = \frac{\beta_{ki}\,\hat{u}_i}{\tau_{ki}}. \tag{7}$$

The index encodes three effects that matter for POP. The factor β_{ki} steers robot k toward nodes that it inspects well, which is the main mechanism that exploits team heterogeneity. The prediction $\hat{u}_i$ accounts for growth during travel, so a distant node with high α_i can outrank a nearby saturated node. Division by travel time converts the score into a rate: the coordinator prefers a nearby high-value target over a distant node that offers only a marginally larger need. A myopic score $\beta_{ki}u_i$ lacks both the arrival prediction and the rate normalization; Section V shows that removing the rate term raises the monitoring cost sharply.

Node-level conflict avoidance alone is not enough when many nodes sit in the same spatial cluster (e.g., a fire focus): several robots can still pick distinct nodes inside one hot region while other regions grow. The coordinator therefore applies a soft regional load-balancing term. If n_i^{near} teammates already claim a location within radius R of node i (by target or current position), the bid becomes

$$\tilde{g}_{ki} = \frac{g_{ki}}{1 + \lambda\, n_i^{\text{near}}}, \tag{8}$$

with R=70 m and λ=2.5 in our experiments. When the mean need inside robot k's own radius-R neighborhood falls below a cool threshold (0.35), an extra factor $(1 + \mu\,\mathbf{1}_{ki}^{\text{loc}})^{-1}$ with μ=1 further discourages camping on that cooled cluster; hot local foci keep their full rate.

Optionally, the coordinator can replace the one-step score by an H-step rollout: it commits to a first node, then greedily continues for $H-1$ further hops on a virtual copy of the need field and accumulates a discounted rate. This paper uses H=1 by default and reports H=3 as an ablation.

B. Sequential-Greedy Auction

Let $\mathcal{F}$ be the set of free robots and $\mathcal{O}$ the set of nodes already committed to by busy robots. While $\mathcal{F}$ is nonempty, the coordinator selects

$$(k^*, i^*) = \arg\max_{k\in\mathcal{F},\, i\notin\mathcal{O}} \tilde{g}_{ki}, \tag{9}$$

assigns i^* to k^*, removes k^* from $\mathcal{F}$, and adds i^* to $\mathcal{O}$ (so later rounds also see the new regional claim). The procedure yields a conflict-free node assignment and, through $\tilde{g}_{ki}$, spreads robots across regions. The underlying greedy matching rule follows the template for monotone submodular maximization on the one-step objective.

Two exclusion rules keep the policy well-behaved. First, a node already inside the deciding robot's reach radius is not a candidate for that robot; this blocks degenerate immediate revisits that would not let need regrow. Second, after any visit the node enters a hard tabu of duration T_{cool}=25 s, shared by the whole fleet, so residual need cannot be farmed in the same round; once the cooldown expires, revisits are allowed again—this is the defining difference between POP and classical OP, where a node is typically visited at most once inside a budgeted tour. In addition, every score is multiplied by a coverage factor $1 + \eta/(1 + n_i^{\text{vis}})$ with η=1.5, which prefers rarely-visited nodes and pushes the fleet to cover more of the graph.

C. Energy Management and Control Loop

Energy sits outside the auction as a hard safety layer. Robot k drains at rate ρ_k. When $e_k \le \underline{e}$ and the robot is not already near a target, the coordinator routes it to the depot x_0, restores $e_k \leftarrow E_k$, and returns the robot to the free pool. Recharge is instantaneous, with no dock queue, and the node is released. This rule couples monitoring with endurance: a policy that chases distant nodes pays more recharge trips and leaves the field unattended for longer.

Algorithm 1 merges growth, visits, recharging, auction, and motion into one tick. The team repeats the tick for the full horizon T. The per-tick cost stays $O(|A|N)$ because each free robot scores all nodes once and the auction performs at most $|\mathcal{F}|$ global argmax passes.

V. Experiments

This section evaluates the reward-rate policy through numerical simulations. A headless kinematic simulator implements the need dynamics, the energy/recharging model, and constant-speed point-to-point motion; all controllers share that layer so the comparison isolates target selection. A 3D Genesis visualization supports qualitative GIS scenes. The study compares four baselines and a matching oracle, then reports ablations, scaling, spatial residual need, a hard stress regime, and two wildfire episodes.

Algorithm 1 Reward-Rate Auction (one control tick)

$u_i \leftarrow \min(u_i + \alpha_i\, dt,\ u_{\max})$ for all i {grow field}
for all robot k **do**
  if k reached its target node **then**
    apply visit: $u_i \leftarrow (1-\beta_{ki})u_i$; mark k free
  end if
  if $e_k \le \underline{e}$ **then**
    route k to the depot
  end if
end for
$\mathcal{F} \leftarrow$ free robots; $\mathcal{O} \leftarrow$ committed nodes
while $\mathcal{F} \neq \emptyset$ **do**
  $(k^*, i^*) \leftarrow \arg\max_{k\in\mathcal{F},\, i\notin\mathcal{O}} g_{ki}$ {Eq. (7)}
  assign i^* to k^*; $\mathcal{F} \leftarrow \mathcal{F}\setminus\{k^*\}$; $\mathcal{O} \leftarrow \mathcal{O}\cup\{i^*\}$
end while
each robot moves toward its assigned node

A. Setup

Each episode lasts T=600 s with step Δt=0.2 s (3000 steps). The simulator draws N nodes uniformly in a 500 m × 500 m area and samples persistence rates $\alpha_i \in [0.005, 0.02]$. The fleet has two groups with small-UAV cruise speeds $(22, 15)$ m/s, energy capacities $E_k \in \{180, 120\}$ (energy units approximate flight-seconds at drain ρ_k=1), and recharge threshold $\underline{e}$=25. In the heterogeneous scenario, β=0.75 on a node's preferred group and β=0.25 otherwise. In the homogeneous scenario, β=0.5 for every pair. Unless noted, each configuration uses N=40, $|A|$=4, and eight random seeds. The primary metric is J (4). Secondary metrics are total collected reward and maximum latency (largest inter-visit gap over nodes). Here $u_{\max}$=1, both groups use ρ_k=1, and the centered depot leaves about 1.4 s of return slack at 15 m/s.

B. Baselines, Oracle, and Cadence

All methods share the same energy, reach, visit cooldown, coverage weight, and motion machinery, so differences come only from target selection. To keep the comparison fair, every policy is evaluated under *two* replan cadences: *online* (replan when an agent is free, or when its current tour ends, without forcing a depot hop) and *depot* (replan only when leaving the depot after start, recharge, or a finished tour return).

Round-Robin. Online: each robot keeps a cyclic pointer and, when free, takes the next feasible index. Depot: at the depot the coordinator commits to an energy-feasible cyclic open tour and executes it until return.

Greedy. Online: free robot k selects $\arg\max_i(\beta_{ki}u_i + w/d_{ki})$ with w=10 [16]. Depot: the same score builds a short energy-feasible open route (≤16 nodes) at each depot departure.

TOP (LKH). The coordinator builds a classical Team Orienteering Problem on the live field (node score $\propto \bar{\beta}_i u_i$, travel budget from energy-feasible range) and solves it with LKH-3 [22]. Depot uses three runs of 400 trials per replan; online uses a lighter call (120 trials) because routes are rebuilt much more often. In both cadences the solver still sees static one-shot scores (no persistence during travel, no per-agent β_{ki}). We assume a shared length budget $\min_k v_k(E_k - \underline{e})/\rho_k$.[1] The LKH trial counts are not equalized across cadences.

GA. A small GA (population 30, 22 generations) builds an energy-feasible route (≤16 nodes) per robot on a shared virtual field [9], under the same online or depot cadence as TOP.

Oracle (matching). We keep g_{ki} and replace the auction by a Hungarian max-weight matching (online) or by joint Hungarian hops that fill a short depot tour (depot). This is optimal for the one-step rate assignment only.

TABLE II
ONLINE CADENCE (REPLAN WHEN FREE). MEAN (± STD) OVER 8 SEEDS, N=40, $|A|$=4. BEST NON-ORACLE ENTRY IN **BOLD**. REWARD/LATMAX AVERAGED OVER BOTH SCENARIOS.

Method	J het ↓	J hom ↓	Reward ↑	LatMax ↓
Round-Robin	36.05 ± 0.44	36.84 ± 0.32	83.1	308.7
GA	30.22 ± 0.75	33.73 ± 0.86	163.6	274.0
TOP (LKH)	31.60 ± 0.67	32.59 ± 0.65	170.5	176.2
Greedy	27.97 ± 0.67	31.50 ± 0.70	223.6	248.6
Reward-Rate (ours)	$\mathbf{25.32 \pm 0.67}$	$\mathbf{29.33 \pm 0.76}$	**243.1**	**136.5**
Oracle (matching)	25.47 ± 1.05	29.08 ± 0.76	244.7	133.9

TABLE III
DEPOT CADENCE (REPLAN ONLY AT DEPOT RETURN). SAME INSTANCES AS TABLE II. BEST NON-ORACLE ENTRY IN **BOLD**.

Method	J het ↓	J hom ↓	Reward ↑	LatMax ↓
Round-Robin	36.61 ± 0.45	37.23 ± 0.46	69.6	597.6
GA	30.69 ± 0.76	33.99 ± 0.61	157.0	401.8
TOP (LKH)	31.35 ± 0.75	$\mathbf{32.88 \pm 0.77}$	166.0	**183.9**
Greedy	30.96 ± 0.75	33.81 ± 0.58	161.6	206.2
Reward-Rate (ours)	$\mathbf{29.72 \pm 0.77}$	32.96 ± 0.66	**177.3**	189.7
Oracle (matching)	29.09 ± 0.55	32.74 ± 0.73	185.7	194.4

Fig. 2 (online cadence) shows the same ordering as the fleet grows from 2 to 8 robots and as N scales from 20 to 80. Fig. 3 visualizes residual need on a shared heterogeneous instance under the online policies.

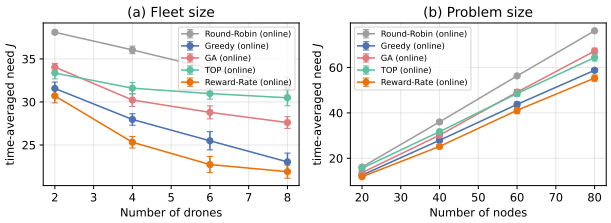


Fig. 2. Scalability of J: (a) fleet size (N=40); (b) nodes ($|A|$=4).

C. Ablation Study

Table IV isolates each design choice on the heterogeneous setting. The coordinator disables, in turn, growth foresight ($\hat{u}_i \leftarrow u_i$), travel-time normalization (score $\beta_{ki}\hat{u}_i$ without dividing by τ_{ki}), the global auction (assign free agents in id order), and the longer rollout (H=3). Removing the rate term raises J by 29.4% and erases most of the gap to Round-Robin; without / τ_{ki} the policy chases distant high-u nodes and spends

[1]For flight speeds $(22, 15)$ m/s and capacities $\{180, 120\}$, the budget is 1425 m, set by the slower group.

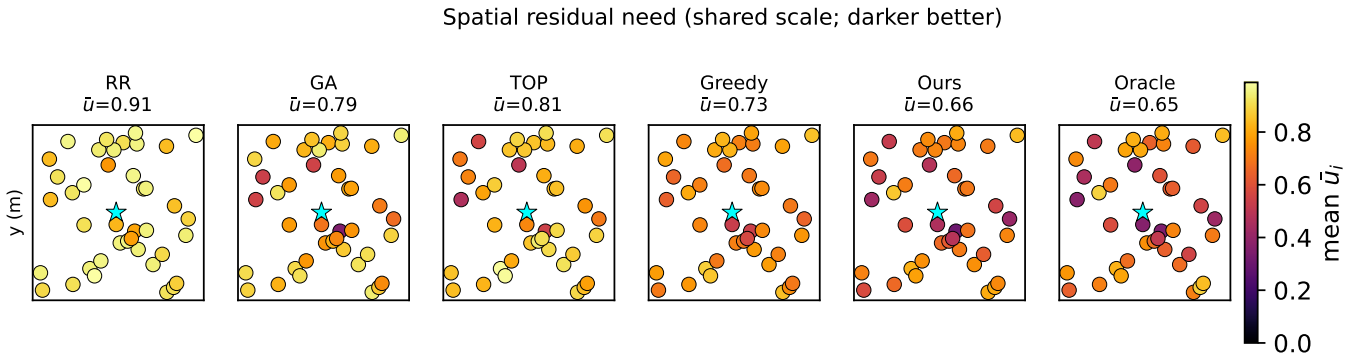


Fig. 3. Spatial distribution of mean residual need $\bar{u}_i$ on a shared heterogeneous instance (common scale; darker is better). $\star$: depot.

TABLE IV
ABLATION OF REWARD-RATE (HETEROGENEOUS, MEAN $\pm$ STD OVER 8 SEEDS).

Variant	$J \downarrow$	Δ vs. full
Full (ours)	$\mathbf{25.32 \pm 0.67}$	—
w/o growth $\hat{u}$	25.54 ± 1.06	$+0.9\%$
w/o rate ($/\,\tau$)	32.76 ± 0.51	$+29.4\%$
w/o auction	25.20 ± 0.63	-0.5%
Rollout $H{=}3$	25.34 ± 0.69	$+0.1\%$

too long in transit. Removing growth foresight increases J by 0.9%, so arrival prediction helps but is secondary at these speeds and densities. Disabling the auction or extending the rollout to $H{=}3$ changes J by less than 1%. In the regimes of this study, the $H{=}1$ auctioned rate index already captures most of the available gain.

D. Hard Stress Test and Wildfire Cases

Table V reports a scarce-coverage regime ($N{=}80$, $|A|{=}3$, faster α, tighter batteries) under both cadences. Absolute J rises for all controllers. Online, Reward-Rate is best among practical methods (2.8% below online TOP). Depot, the rate index still slightly undercuts depot TOP (-0.5%). There the one-step oracle is lower (75.52 versus 75.79). Capacity (not cadence alone) sets a high floor on J, but the rate policy allocates the scarce visits more effectively in both blocks.

TABLE V
HARD STRESS TEST ($N{=}80$, $|A|{=}3$, 6 SEEDS). Δ RELATIVE TO TOP IN THE SAME CADENCE.

	Online		Depot	
Method	J	Δ	J	Δ
Round-Robin	79.19 ± 0.07	$+4.1\%$	78.81 ± 0.17	$+3.5\%$
GA	78.47 ± 0.13	$+3.2\%$	78.32 ± 0.15	$+2.8\%$
TOP (LKH)	76.04 ± 0.50	—	76.17 ± 0.40	—
Greedy	75.07 ± 0.33	-1.3%	76.86 ± 0.42	$+0.9\%$
Reward-Rate	$\mathbf{73.92 \pm 0.53}$	$\mathbf{-2.8\%}$	$\mathbf{75.79 \pm 0.37}$	$\mathbf{-0.5\%}$
Oracle	73.91 ± 0.65	-2.8%	75.52 ± 0.53	-0.8%

Fig. 4 shows two GIS wildfire episodes on the same orthophoto. The demonstration uses five staggered foci and four drones; the hard scene uses eight foci and three drones. As foci ignite, local need rises until a visit reduces it, and the auction spreads agents across fronts according to β_{ki}.

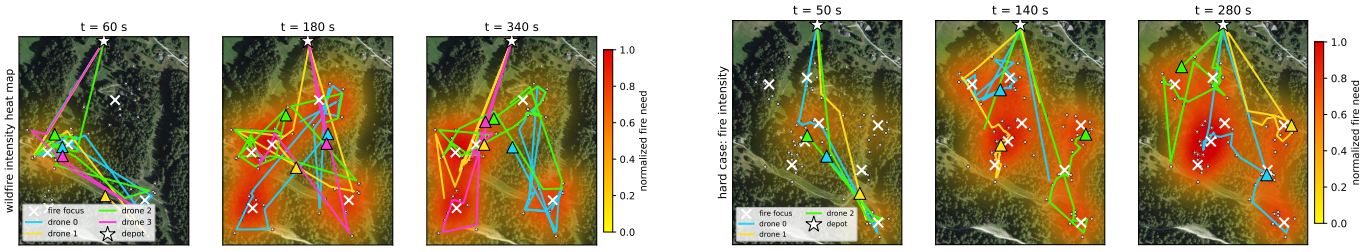
E. Results and Discussion

Tables II–III and Fig. 5 separate the two cadences so that index design is not confounded with replan frequency. Under the *online* cadence, Reward-Rate reduces J by 19.9% (heterogeneous) and 10.0% (homogeneous) relative to online TOP, and by 9.5% / 6.9% relative to online Greedy. The matching oracle stays within seed noise of Reward-Rate. Under the *depot*, the same rate index still leads among practical methods on the heterogeneous setting (5.2% below depot TOP), while on the homogeneous setting it ties depot TOP within noise ($+0.2\%$). Absolute J rises for every method when forced to commit to full sorties, which shows that much of the online gap is reactivity.

Round-Robin ignores u_i, α_i, and β_{ki}. Classical TOP selects a high-score subset under a travel budget, yet its static scores and averaged β cannot encode persistence or agent-specific quality; making TOP online (lighter LKH calls whenever a tour ends) does not close the gap to Reward-Rate because the objective inside LKH remains one-shot. The GA shares TOP's cadence in each block and trails the rate policy. Greedy reacts quickly online but without $/\,\tau$ still over-commits to distant bright nodes. Visit cooldown and coverage apply uniformly, so within each table the remaining gap is attributable to scoring and assignment.

TOP latency moves from 176.2 to 183.9 across cadences; Reward-Rate from 136.5 to 189.7. Robots in one group share energy and drain, so they tend to recharge together. On heterogeneous instances the reduction versus TOP falls from 19.9% online to 5.2% at the depot: matched cadence removes about three quarters of that gap. The rest mixes the rate index with one-shot mean-quality scores, not J, and a closed depot tour against open-field replanning. On homogeneous instances the online 10% becomes the depot tie. Online LKH uses 120 trials per call and depot LKH 1200; calls per episode were not logged.

These results support three conclusions. First, scoring visits as a growth-aware rate is the main lever: the ablation (Table IV) shows that removing the $/\,\tau_{ki}$ term raises J by 29.4%, collapsing toward Round-Robin. Second, matched online and depot cadences reveal that reactivity explains much of the online gap to TOP, while a heterogeneous depot residual (5%) remains. Third, making classical TOP online does not



(a) Demonstration. Video: `https://youtu.be/OoaR1NY2Fes`.

(b) Hard stress. Video: `https://youtu.be/htKDRWT-_qQ`.

Fig. 4. GIS wildfire cases at three instants. Curves/triangles: routes/positions; ⋆: depot.

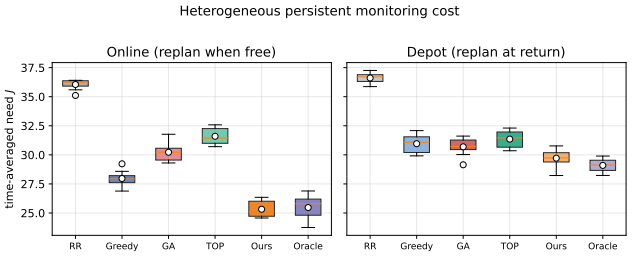


Fig. 5. Time-averaged cost J on the heterogeneous setting (lower better). Left: online cadence; right: depot cadence. Labels: RR, Greedy, TOP, GA, Ours, Oracle.

close that gap, because the solver's one-shot scores remain misaligned with the persistent objective.

Table V reports the scarce-coverage regime (N=80, $|A|$=3): online, Reward-Rate is best among practical methods (2.8% below online TOP); depot, the rate index slightly undercuts depot TOP (-0.5%). Capacity, not cadence alone, sets a high floor on J, but the rate policy allocates scarce visits more effectively in both blocks. Fig. 4 shows two GIS wildfire episodes where the auction spreads agents across fronts according to β_{ki} as foci ignite and need regrows.

This study has clear limitations. The simulator uses kinematic point robots, perfect communication, and a central coordinator. Need growth is independent across nodes and linear in time; the model does not yet include wind-driven fire spread, sensor noise, or localization error. The matching oracle is a one-step reference on g_{ki}, not a global lower bound on the infinite-horizon POP.

VI. Conclusion

This paper introduced the Persistent Orienteering Problem and a centralized reward-rate auction for heterogeneous multi-robot persistent monitoring. Under matched online and depot replan cadences, the policy reduces J by about 10–20% versus online TOP and by about 5% versus depot TOP on heterogeneous instances, and stays close to the one-step matching oracle. Under scarce coverage (N=80, three robots), that reduction falls to 2.8% online and 0.5% at the depot. Future work includes noisy observations, intermittent communication, robot failures, spatially correlated need fields, longer-horizon rollouts with guarantees, decentralized variants, and hardware validation under real dynamics. Also open are a time-varying baseline, a TOP with persistence and per-robot quality, turning constraints, distinct drain rates, and a sweep of the regional weights.

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